for (0 → N) → O(N) × O(1)
= O(N)

$O(N^2) + O(N)$

$O(N^2) + O(n) + O(1)$

discard lower orders.

2) O(1)
O(N)
O(N) → $O(N^2)$

3) a) $\Theta N - K$.
$= \Theta N - \cancel{KN}$
$= \Theta N = O(N)$

coeff ignore

lower term ||

b) $N > \log N$

$O(N + 8\log N)$

$O(N)$

d) $O(N + M^2)$, N & M both input

$= O(N + M^2)$

(6)

```
for (x=1; x<=n; x*=2)
    print x;
```

3

n=10

↳ x=1

↳ 2

$O(\log n)$

↳ 4 ∫ 4 times → 2
↳ 8 3
↳ 16 ✗

n = 13	14	15	16	18
run → 4	4	4	5	5

$\log_2 8 = 3$

$\log$

(7) $a = 0, \; i = N$

```
while (i > 0) {
    a += i;
    i /= 2;
}
```

$i = 2$

8 4 2 1 $O(\log_2(N))$

8	12	14	13

6

0 → 2 → 4 →

i++
i+=2
i+=4
} O(n)

i = 0 to N, i*=2 → O(log N)

i/=2 → O(log n)

(10) Which one ≠ O(n²)?

(10) $2^{n+1} + 3(n+1)!$

$\Rightarrow$ a) ~~$(15^{10})\ n + 1209$~~

$\Rightarrow O(n) = O(n^2)$

b) $n^{1.98} = O(n^2)$

c) $n^3/\sqrt{n} = \frac{n^3}{n^{1/2}} = n^{3-\frac{1}{2}}$

$= n^{5/2} = n^{2.5}$

$\neq n^2$

d) $(2^{2v}) * n$

$= O(n) = O(n^2)$